Appendix B: UAV/UGV Part List

The following table is an example part list to build a mini UAV/UGV. The total cost of all vehicles from each team that participates in this event must not have a cost exceeding $350 (Excluding Tax). This build ends up with a net cost of approximately $300 (not including a RC radio transmitter/receiver, small parts such as screws, hot glue, etc.). Students will need to devise a carry and release system for the UGV in addition to the parts listed here. The carry and release system will not be considered when evaluating the budget for each UAV and UGV.

Unmanned Aerial Vehicle (UAV) Build Example					
Component	**Example Build**	**Vendor**	**Qty**	**Unit Cost**	**Net Cost**
Frame	Design and Fabricate	-	-	-	-
Flight Controller*	F4 (Base Model)	getFPV	1*	$26.99	$26.99
	HyperLite F4	getFPV	1	$32.99	$32.99
Motor	Turnigy D2830-11 1000kv Motor	Hobbyking	4	$11.26	$45.04
Speed Controllers*	Turnigy MultiStar 30A BLHeli-S	Hobbyking	4	$10.56	$42.24
	BL-Heli 32bit 30AMP ESC, No UBEC	Amazon	1	$41.99	$41.99
	RaceStar 35A 4in1 ESC	getFPV	1	$31.99	$31.99
Propellers	Gemfan 8045 (8x4.5) Propellers	Amazon	1	$8.99	$8.99
Battery	ZIPPY Compact 3000mAh 4S1P 20C	Hobbyking	1	$21.30	$21.30
Power Distribution	ESC Power Breakout Cable	Amazon	1	$11.99	$11.99
Battery Alarm	Hobbyking Low Voltage Alarm	Hobbyking	1	$2.81	$2.81
Battery Strap	Turnigy Battery Strap 330mm	Pyramidmodels	1	$2.50	$2.50
Unmanned Ground Vehicle (UGV) Build Example					
Frame / Structural Components	Design and Fabricate	-	-	-	-
Microcontroller	Arduino Teensy	Amazon	1	$26.66	$26.66
Color Sensor	TCS230 TCS3200 Color Module	Amazon	1	$5.99	$5.99
Servo	Micro Servo - High Torque	Adafruit	1	$9.95	$9.95
Motors	25:1 Metal Gearmotor	Polulu	2	$22.95	$45.90
Motor Brackets	Polulu 20D mm Metal Gearmotor Brackets	Polulu	1	$6.95	$6.95
Motor Driver	TB67H420FTG Dual Motor Driver	Polulu	1	$9.95	$9.95
IR Sensor	QTR-8A IR Sensor	Polulu	1	$9.95	$9.95
5V / 12V regulator	LM2596 Voltage Regulator	Amazon	1	$11.75	$11.75
Battery	Floureon 7.4V 1000mAh Battery	Amazon	1	$13.99	$13.99
Breadboard	400 pin 4 power rail	Amazon	1	$6.99	$6.99
Total Cost		**$299.69**			

Note: Part types labeled with a * have multiple parts listed. Require only one of the listed part to build a complete design. The example build was constructed using the least expensive parts listed items.